TUH preprocessing analysis — is the LaBraM protocol right for TUEG and TUAB?

LaBraM, BIOT and CBraMod prepare TUAB with one fixed recipe: keep 23 referential channels, band-pass 0.1–75 Hz, **notch 50 Hz**, resample to 200 Hz, express in µV (`dataset_maker/make_TUAB.py`, walked through step by step in `notebooks/tuab_data_preparation.ipynb`). TUH is recorded in the US, where mains power is **60 Hz**, and the full TUEG corpus mixes sampling rates (250–1000 Hz), equipment and two decades of recordings. Before converting 1.76 TB of TUEG this notebook measures, on a stratified sample of real recordings, what the protocol leaves in the signal and whether any of it could act as a shortcut for brain age.

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The measurements come from `labram/eval/preprocessing_qc.py` (`analyze_recording`), run on a 120 s segment (60–180 s after the start) of each recording.

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187 recordings analysed, 0 errors
```

1. The protocol and the variants tested

For a channel $x(t)$ in µV sampled at f_s Hz, the LaBraM protocol computes

$$y = \mathrm{resample}_{200\,\mathrm{Hz}}\left(\mathrm{notch}_{50}\left(\mathrm{bp}_{[0.1,\,75]}(x)\right)\right) \tag{1.1}$$

where $\mathrm{bp}_{[0.1,75]}$ is MNE's zero-phase FIR band-pass (`raw.filter(0.1, 75)`, transition bands chosen automatically: 0.1 Hz below, $\min(\max(0.25 \cdot 75, 2), f_s/2 - 75)$ Hz above), notch_{f_0} is MNE's FIR notch at f_0 Hz (`raw.notch_filter(f0)`), and $\mathrm{resample}_{200}$ is MNE's FFT resampler (`raw.resample(200)`). Table 1.1 lists the variants measured next to it; each changes only the notch or the low-pass of Eq. (1.1) (`labram/eval/preprocessing_qc.py::PROTOCOLS`).

Table 1.1. Protocol variants

One row per variant: notch frequencies (Hz), band-pass edges (Hz), output rate (Hz).

protocol	notch (Hz)	band-pass (Hz)	output rate (Hz)
labram (50 Hz notch)	50	0.1–75	200.00
60 Hz notch	60	0.1–75	200.00
50 + 60 Hz notch	50, 60	0.1–75	200.00
60 Hz notch, 45 Hz low-pass	60	0.1–45	200.00

2. The sample

TUEG recordings were drawn from the corpus index (`dataset_maker/tuh_index.py`), restricted to recordings with all 23 channels, a known age and at least 5 minutes, **one recording per patient**, up to 12 per cell of native sampling rate × recording year. TUAB adds 40 normal and 40 abnormal recordings (one per patient). Table 2.1 gives the counts.

The recording year is the year in the EDF header's `Startdate` field (the date the EEG was recorded, not the patient's age), grouped as ≤2004, 2005–10 and 2011+. Sampling rate and recording year are not independent in TUEG: 400 Hz is almost only ≤2004, 512 Hz and 1000 Hz almost only 2011+, and 256 Hz never appears in 2005–10 (empty cells in Table 2.1). Any property of the equipment therefore also carries the recording date — and through it, as Section 6 shows, the patient's age.

Table 2.1. Sample composition

Recordings per group × native sampling rate (Hz) × recording year (from the EDF header).

group	recording year sfreq	≤2004	2005–10	2011+	total
TUAB	**250**	0	14	62	76
	256	0	0	4	4
	400	0	0	0	0
	512	0	0	0	0
	1000	0	0	0	0
TUEG	**250**	12	12	12	36
	256	12	0	12	24
	400	12	5	1	18
	512	2	0	12	14
	1000	3	0	12	15

Table 2.2 shows the EDF header fields that could say how a recording was already filtered or scaled: the physical unit of the 23 channels and the free-text *prefiltering* field. Every recording is in µV, and the prefiltering field carries no information (`HP:0 LP:0 N:0` , or `-1` placeholders) — the headers cannot tell us whether a hardware notch was applied, so mains interference has to be measured in the signal itself (Section 4).

Table 2.2. EDF header unit and prefiltering fields

Recordings per (unit, prefiltering text, group).

	group	TUAB	TUEG
units	**prefilter**		
uV	**HP:-1.000 Hz LP:-2.0 Hz N:0.0**	0	30
	HP:0.000 Hz LP:0.0 Hz N:0.0	80	77

3. Measures

Spectrum. For a segment $x_c(t)$, $c = 1..C$ channels, the channel-median power spectral density is

$$P(f) = \text{median}_c \;\; \text{Welch}\left(x_c\right)(f) \qquad [\mu\text{V}^2/\text{Hz}] \tag{3.1}$$

with 4 s Hann segments and 50% overlap (0.25 Hz resolution; `preprocessing_qc.py::channel_median_psd`). The median over channels keeps one bad electrode from dominating.

Line-noise prominence. How far a narrow band around f_0 stands above its neighbourhood:

$$\text{LNP}(f_0) = 10 \log_{10} \frac{\text{mean}_{|f-f_0|\le 0.5}\, P(f)}{\text{median}_{2\le|f-f_0|\le 5}\, P(f)} \qquad [\text{dB}] \tag{3.2}$$

(`line_prominence_db`). $\text{LNP} \approx 0$ dB means no line; 10 dB means a line ten times its surroundings; a negative value is a dip, e.g. the hole a notch filter cuts. It is measured at the mains fundamentals (50, 60 Hz), the 2nd/3rd US harmonics (120, 180 Hz), and where harmonics would fold back by aliasing: 180 Hz sampled at 250 or 256 Hz lands at 70 or 76 Hz, and after resampling to 200 Hz, 120 Hz lands at 80 Hz and 180 Hz at 20 Hz.

Mains share of the output. The share of the 0.5–75 Hz power that falls within ±5 Hz of 60 Hz:

$$F_{55\text{–}65} = \frac{\sum_{55 \le f \le 65} P(f)}{\sum_{0.5 \le f \le 75} P(f)} \tag{3.3}$$

(`band_fraction`).

Amplitude and integrity (`amplitude_stats` , on the raw 23-channel segment in µV, T samples):

$$A_{50} = \mathrm{median}_{c,t}\,|x_c(t)|, \qquad A_{99} = \mathrm{quantile}_{0.99,\,c,t}\,|x_c(t)| \qquad [\mu\mathrm{V}] \tag{3.4}$$

$$\mathrm{clip} = \frac{1}{CT} \sum_{c,t} 1\big[\,|x_c(t)| \ge 0.995\, x_{\max}\big] \tag{3.5}$$

where $x_{\max}$ is the EDF physical maximum (±5000 µV for TUH), and

$$\mathrm{flat} = \frac{1}{C} \sum_{c} 1\Big[\min_{k} \mathrm{std}\left(x_c[\text{10 s window } k]\right) < 0.5\,\mu\mathrm{V}\Big] \tag{3.6}$$

is the share of channels that are flat (disconnected or saturated) in some 10 s window.

4. Mains interference in the raw recordings

Figure 4.1 averages $\log_{10} P(f)$ of Eq. (3.1) over the recordings of each sampling-rate group, 40–130 Hz (up to each group's Nyquist frequency). Table 4.1 gives the median LNP of Eq. (3.2) per group and the share of recordings with a clear line ($\mathrm{LNP}_{60} > 6$ dB, a 4× peak).

Figure 4.1. Raw spectrum around the mains frequencies (before any filtering)

(a) TUEG and (b) TUAB: mean over recordings of $\log_{10} P(f)$ (µV²/Hz, Eq. 3.1) of the raw 120 s segment **before any filtering** (native sampling rate, no band-pass, no notch), one line per native sampling rate; dashed lines at 50 and 60 Hz. n per line in Table 2.1. The curves are *means* of log-spectra, so a minority of recordings with strong lines produces a visible peak even where the *median* recording has almost none (Table 4.1).

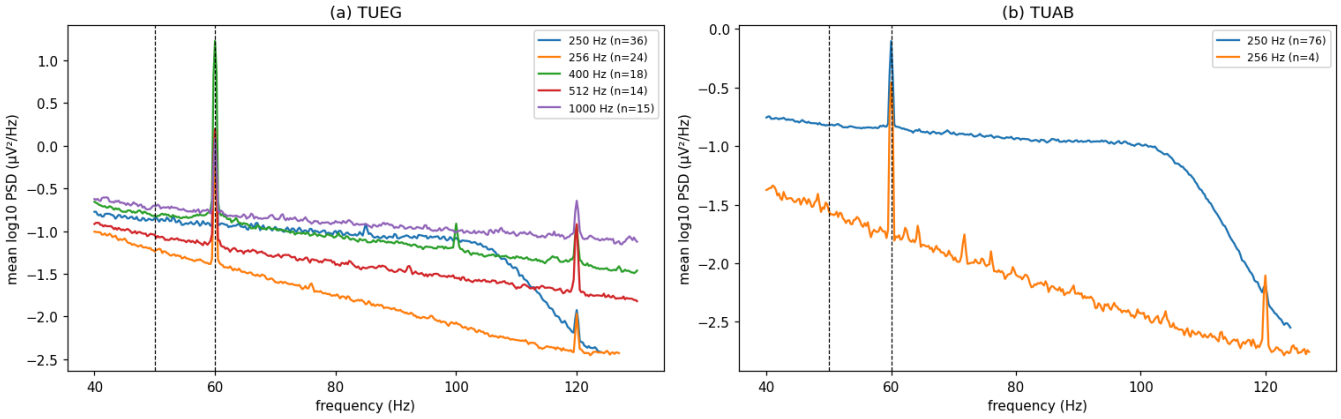


Table 4.1. Line-noise prominence of the raw recordings (before any filtering)

Median LNP (dB, Eq. 3.2) of the unfiltered recordings at each probe frequency per group × native sampling rate; `Hz` = native sampling rate; `60 p75` / `60 p90` = 75th/90th percentile of LNP_{60}; `%>6dB` / `%>20dB` = percentage of recordings with LNP_{60} above 6 / 20 dB; `n` recordings. – where the frequency is above the group's Nyquist limit (half the sampling rate).

group	Hz	50 Hz	60 Hz	120 Hz	180 Hz	70 Hz	76 Hz	60 p75	60 p90	%>6dB	%>20dB	n
TUAB	**250**	0.1	1.9	-0.4	–	0.0	0.1	5.7	12.7	21	8	76
	256	0.2	3.3	0.8	–	-0.3	0.1	11.9	24.7	25	25	4
TUEG	**250**	-0.0	4.5	0.1	–	0.0	0.1	9.1	20.4	36	11	36
	256	-0.1	5.4	1.4	–	0.0	0.1	12.1	18.9	46	8	24
	400	0.0	14.8	1.1	3.6	0.1	-0.0	33.8	40.2	56	44	18
	512	0.0	5.9	3.9	5.3	-0.0	-0.2	14.2	23.8	50	14	14
	1000	0.1	2.8	0.1	0.6	0.2	-0.0	5.3	16.9	20	7	15

Table 4.1 and Figure 4.1 give the first answer. **There is no 50 Hz line anywhere** (median $\mathrm{LNP}_{50} \approx 0$ dB in every group), so the protocol's 50 Hz notch removes no interference — it only cuts a hole in genuine EEG at 50 Hz (Section 5). **A 60 Hz line is present in about half of the recordings** (`%>6dB` per group in Table 4.1; over the whole sample, a third exceed 6 dB), and its strength depends on the equipment: weakest in TUAB and in TUEG's 1000 Hz recordings, strongest in the 400 Hz group (the ≤2004 systems of Table 2.1). The 120 Hz and 180 Hz harmonics are visible where the sampling rate allows them (above 240 / 360 Hz native). The acquisition-alias probes (70, 76 Hz) are flat: the 180 Hz harmonic is not folded into the band.

5. What each protocol leaves at the model input

Every recording was passed through the four protocols of Table 1.1 and Eqs. (3.2)–(3.3) were measured on the 200 Hz output. Figure 5.1 compares the output spectra of the LaBraM protocol and the 60 Hz-notch variant; Table 5.1 summarises all four.

Figure 5.1. Model-input spectrum around 60 Hz after processing, LaBraM vs 60 Hz notch

Mean over recordings of $\log_{10} P(f)$ (µV²/Hz, Eq. 3.1) of the 200 Hz output **after processing**, 40–90 Hz, one line per native sampling rate. Left column: LaBraM protocol (50 Hz notch); right column: the same with a 60 Hz notch instead. (a, b) TUEG (n = 107), (c, d) TUAB (n = 80). Dashed lines at 50, 60 and 75 Hz (low-pass edge).

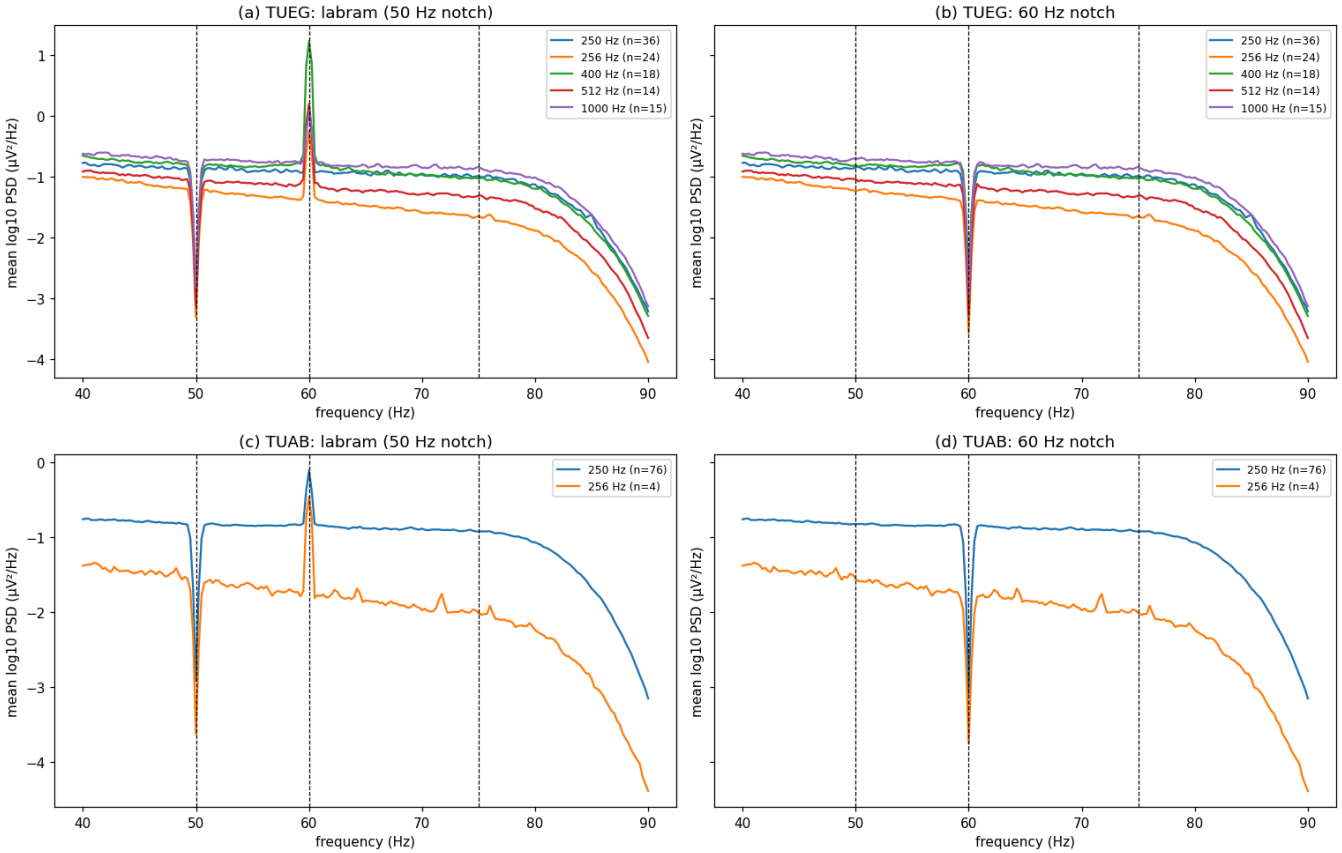


Table 5.1. Line noise left at the model input after processing, per protocol

TUEG + TUAB, all 187 recordings, after processing: median LNP (dB, Eq. 3.2) at 50 and 60 Hz and at the resampling-alias probes 20 and 80 Hz; `share60>6dB` = share of recordings with a 60 Hz line above 6 dB left; `F55-65` = median mains share of the output power (Eq. 3.3, %).

protocol	50 Hz	60 Hz	20 Hz (alias of 180)	80 Hz (alias of 120)	share60>6dB	F55-65 (%)
labram (50 Hz notch)	-5.05	3.13	0.05	0.46	0.33	1.39
60 Hz notch	0.04	-5.35	0.05	0.46	0.01	0.83
50 + 60 Hz notch	-5.05	-5.35	0.05	0.46	0.01	0.83
60 Hz notch, 45 Hz low-pass	-0.42	-5.68	0.06	-0.99	0.01	0.00

Table 5.1 is the core result. The LaBraM protocol passes the 60 Hz line through unchanged (its median LNP_{60} equals the raw value of Table 4.1) and cuts a dip at 50 Hz where there was no interference. A 60 Hz notch removes the line in almost every recording (median LNP_{60} becomes a dip of about −5 dB, the notch itself; `share60>6dB` drops to about 1%, the few recordings whose line is wider than MNE's default notch width of $f_0/200$ Hz). Notching both 50 and 60 Hz only adds the needless 50 Hz dip. Resampling causes no systematic aliasing (median of the 20 and 80 Hz probes ≈0 dB): MNE's FIR low-pass at 75 Hz attenuates 120/180 Hz before the 200 Hz resampler; the 80 Hz probe lies in the low-pass transition band, where the little that remains is already strongly attenuated. A 45 Hz low-pass removes the mains band entirely but also discards 45–75 Hz gamma/EMG content that the pre-trained LaBraM tokenizer was trained on.

6. Is the residual a bias?

A 60 Hz line matters for brain age only if its strength is related to age. Figure 6.1 plots the raw LNP_{60} against age and by recording year; Table 6.1 gives Spearman correlations

$$\rho_S(u, v) = \mathrm{corr}_{\mathrm{Pearson}}\left(\mathrm{rank}(u), \mathrm{rank}(v)\right) \tag{6.1}$$

(`scipy.stats.spearmanr` , two-sided p), between LNP_{60} and age, and between age and the native sampling rate, per group.

Figure 6.1. 60 Hz line strength vs age and recording year

(a) Raw LNP_{60} (dB, Eq. 3.2) against age (years), TUEG (dots, colour = native sampling rate) and TUAB (crosses); (b) raw LNP_{60} by recording year, TUEG. n in Table 2.1.

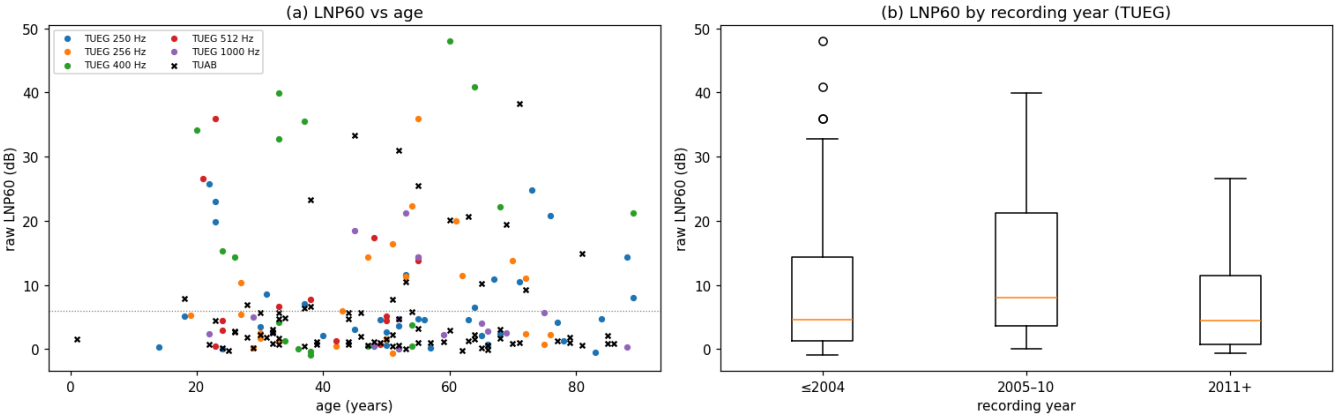


Table 6.1. Correlation of line strength and equipment with age

Spearman ρ_S (Eq. 6.1) and two-sided p per group; `n` recordings.

group	variable vs age	rho_S	p	n
TUAB	**raw LNP60**	-0.026	0.821	80
	native sampling rate	-0.193	0.087	80
	recording year	-0.154	0.173	80
TUEG	**raw LNP60**	0.008	0.936	107
	native sampling rate	-0.148	0.127	107
	recording year	0.098	0.314	107

Table 6.2. Patient age by equipment and recording year across the TUEG corpus

All TUEG recordings of the corpus index with a usable age (not only the sample): mean and median patient age (years) and `n` recordings per native sampling rate and per recording year.

		mean age	median age	n
sampling rate (Hz)	**250**	52.3	55.0	22443
	256	50.5	53.0	34153
	400	46.9	47.0	3057
	512	46.6	46.0	1510
	1000	52.4	54.0	3062
recording year	**≤2004**	45.1	47.0	6089
	2005–10	52.7	55.0	11180
	2011+	51.3	53.0	46643

Within the sample, line strength is **not** directly correlated with age (Table 6.1, $\rho_S \approx 0$ in both groups): the 60 Hz line is not a recording-level age marker. The risk is at group level. Table 6.2 shows on the whole corpus that patients recorded on different equipment generations, and in different recording years, differ in mean age by several years — the 400 Hz and 512 Hz systems and the ≤2004 recordings are the youngest — and Table 4.1 shows that the 400 Hz (≤2004) systems also carry the strongest 60 Hz line. A model can pick up such equipment fingerprints and use them as a proxy for the age mix of the sub-population they come from. Removing the line (Table 5.1) cuts this path at no cost: brain-age models use no physiological signal in a 1 Hz band at 60 Hz. Equipment and recording year should in any case be reported as strata of the evaluation (error per sampling rate and recording year), since other fingerprints than the line remain.

7. Amplitude scale and signal integrity

Table 7.1 gives the amplitude and integrity measures of Eqs. (3.4)–(3.6) per group and, for TUAB, per normal/abnormal label. Figure 7.1 plots A_{50} against age and by sampling rate.

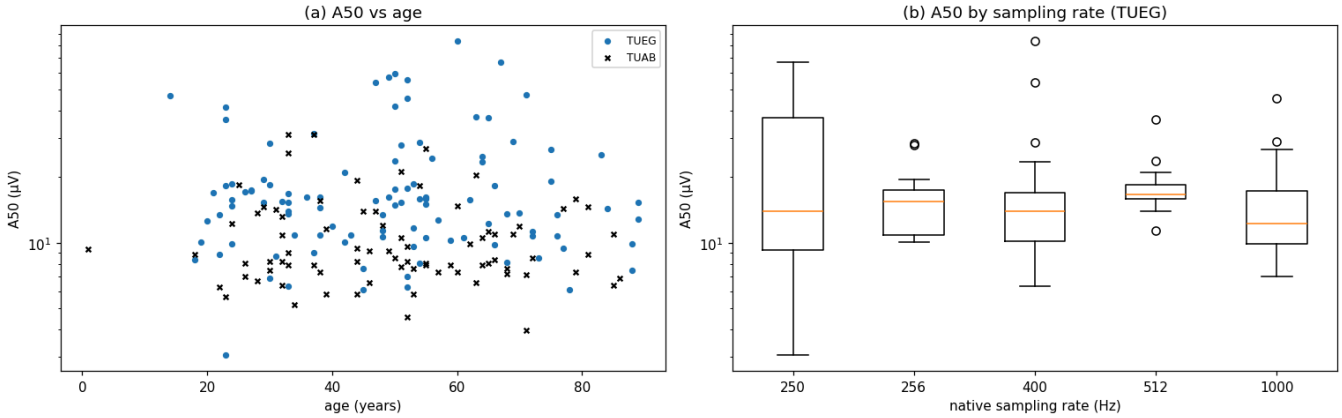
Table 7.1. Amplitude and integrity of the raw signal

Medians over recordings of A_{50}, A_{99} (µV, Eq. 3.4), clip share (Eq. 3.5) and flat-channel share (Eq. 3.6); `n` recordings.

cohort	A50 (µV)	A99 (µV)	clip	flat	n
TUAB abnormal	9.915	77.476	0.000	0.000	40
TUAB normal	8.469	80.910	0.000	0.000	40
TUEG	15.262	124.515	0.000	0.000	107

Figure 7.1. Signal amplitude vs age and sampling rate

(a) A_{50} (µV, Eq. 3.4, log scale) against age (years) for TUEG (dots) and TUAB (crosses); (b) A_{50} by native sampling rate, TUEG. n in Table 2.1.



No recording in the sample clips or has a flat channel (Table 7.1), so integrity filtering is a corpus-level concern (missing channels, very short files; corpus index), not a per-sample one. Amplitude is a different matter: TUEG's typical amplitude is markedly higher than TUAB's (Table 7.1) — TUEG is not curated to routine recordings the way TUAB is — and it varies between sampling-rate groups (Figure 7.1(b)), i.e. between equipment generations. In this sample A_{50} shows no clear trend with age (Figure 7.1(a)), so scale is mainly a corpus/equipment difference here. LaBraM feeds the model $x/100$ (fixed scaling, `labram/train/train_finetune.py`), which keeps both. Per-patch z-scoring (`labram_plus.z_score`, Section 9 of `notebooks/tuab_data_preparation.ipynb`) removes absolute scale entirely; common average reference (`labram_plus.common_average_reference`, on in the brain-age config) removes the reference-dependent part shared across channels. Which of the two to use for TUEG is an empirical question for the training experiments (Section 9).

8. Removing the line without a hole

A notch filter removes a band around 60 Hz whether or not there is interference in it: on a recording without a line it cuts a hole into genuine EEG (the dips in Figure 5.1). Two alternatives remove only the interference.

Sinusoid fitting (MNE `notch_filter(method="spectrum_fit")`, the approach of EEGLAB's CleanLine): in sliding 10 s windows, a 60 Hz sinusoid is estimated by multitaper regression and subtracted,

$$\hat{x}(t) = x(t) - \hat{a}\cos(2\pi\,60\,t) - \hat{b}\sin(2\pi\,60\,t) \qquad (8.1)$$

where $\hat{a}, \hat{b}$ (µV) are the window's fitted amplitudes, kept only when a multitaper F-test finds the line significant.

Spectrum interpolation (Leske & Dalal 2019; `preprocessing_qc.py::spectrum_interpolate`): the short-time Fourier transform $Z(f,\tau)$ with 10 s Hann windows and 50% overlap is modified in the band $|f-60| \le h$ as

$$Z'(f,\tau) = \sqrt{\frac{\mathrm{median}_{h<|f'-60|\le h+3}|Z(f',\tau)|^2}{\ln 2}}\; e^{i\angle Z(f,\tau)}, \qquad |f-60| \le h, \quad (8.2)$$

i.e. the power in the band is set to the power level of the 3 Hz on either side and the phase is kept; the inverse transform gives the cleaned signal. h = 1.5 Hz. Away from the band the transform is an exact round trip. The flank level is the median power divided by $\ln 2$: for noise-like EEG the power of a frequency bin is exponentially distributed, whose median is $\ln 2$ times its mean, so this estimates the mean power without bias, while an unrelated narrow peak inside the flanks (one recording of the sample has one at 53 Hz) cannot pull it up. (Using the mean *magnitude*, as a first version did, is biased low by a factor $4/\pi$ in power, about 1 dB, and cuts a shallow hole where there is no line.)

Both are compared with the 60 Hz notch on two subsets of the sample: recordings **with a strong line** (raw $\mathrm{LNP}_{60} > 10$ dB), where the question is how much of the line is left, and **clean** recordings (raw $|\mathrm{LNP}_{60}| < 1$ dB), where the question is how deep a hole the method cuts. Table 8.1 gives the numbers, Figure 8.1 the average spectra, and Figures 8.2–8.6 single recordings as EEG traces and spectrograms.

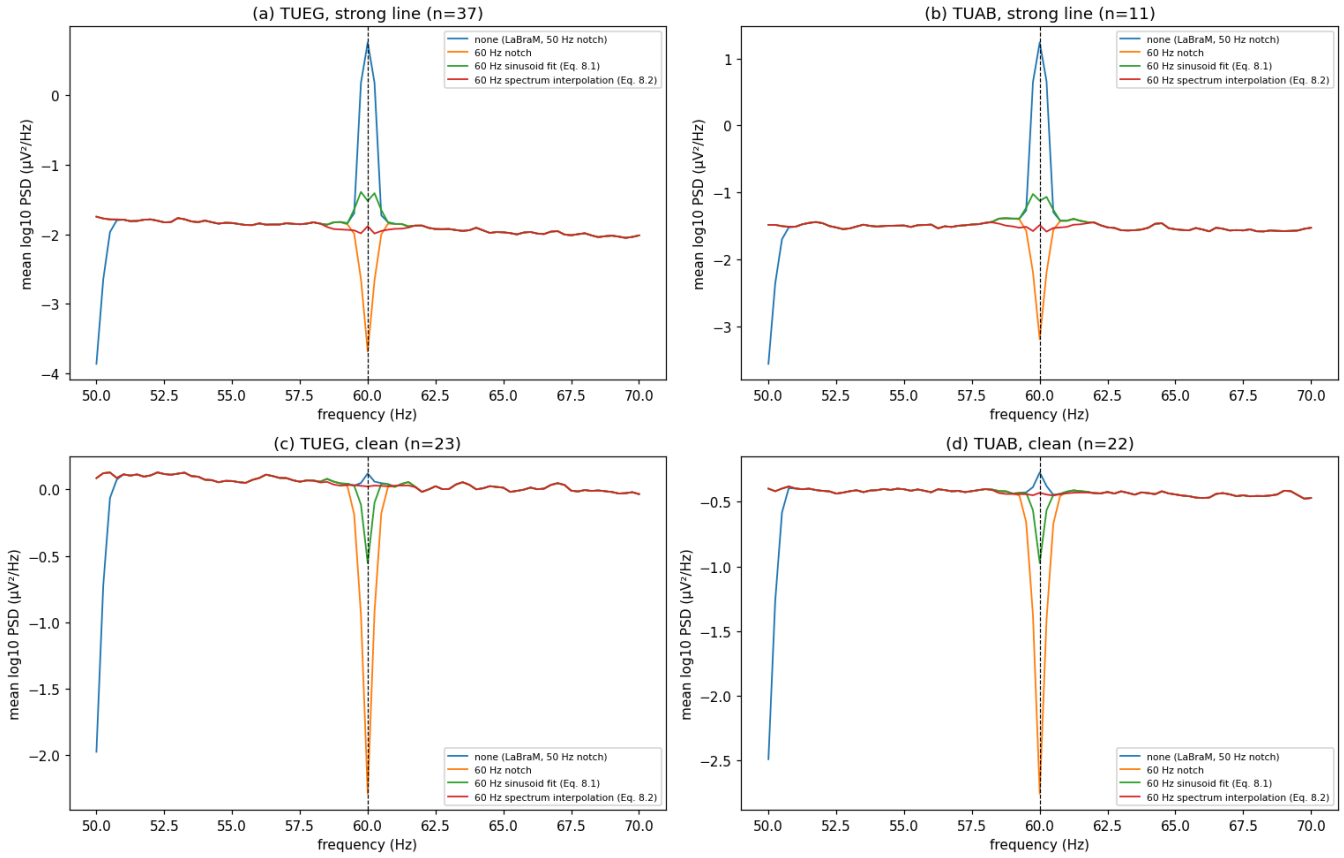
Table 8.1. Line left and hole cut by each line-removal method (after processing)

LNP_{60} (dB, Eq. 3.2) of the 200 Hz output after band-pass, the line-removal method and resampling. Strong-line recordings (raw $\mathrm{LNP}_{60} > 10$ dB): median and maximum line left (lower is better, 0 dB = no line). Clean recordings (raw $|\mathrm{LNP}_{60}| < 1$ dB): median LNP_{60}, where a negative value is the hole the method cuts (closer to 0 is better). `n` recordings per subset; TUEG + TUAB.

method	strong: median left (dB)	strong: max left (dB)	clean: median (dB)	clean: min (dB)	n strong / clean
none (LaBraM, 50 Hz notch)	20.0	48.0	0.4	-1.0	48 / 45
60 Hz notch	-4.8	7.4	-5.6	-6.9	48 / 45
60 Hz sinusoid fit (Eq. 8.1)	2.0	26.6	-1.3	-2.7	48 / 45
60 Hz spectrum interpolation (Eq. 8.2)	-0.5	2.2	-0.2	-0.9	48 / 45

Figure 8.1. Spectrum around 60 Hz after each line-removal method

Mean over recordings of $\log_{10} P(f)$ (μV²/Hz, Eq. 3.1) of the 200 Hz output after processing, 50–70 Hz, one line per method of Table 8.1: (a) TUEG and (b) TUAB recordings with a strong line (raw $\mathrm{LNP}_{60} > 10$ dB); (c) TUEG and (d) TUAB clean recordings (raw $|\mathrm{LNP}_{60}| < 1$ dB). n in each panel title. A good method is flat at 60 Hz in all four panels: no peak in (a, b), no dip in (c, d).



Spectrum interpolation removes the line best and cuts no hole. On the strong-line recordings it leaves no line (Table 8.1: median about −0.5 dB, worst case about 2 dB — better than the notch's worst case of 7.4 dB, whose fixed narrow band misses lines that are broader than it), and on clean recordings it changes the 60 Hz band by only a few tenths of a dB, against a hole of about 5.6 dB from the notch (Figure 8.1(c, d)). Sinusoid fitting sits in between: on clean recordings it still cuts a narrow dip exactly at 60 Hz (shallower than the notch's, Figure 8.1(c, d)), and on strong-line recordings it misses part of the line (worst case above 25 dB, Figure 8.1(a, b)) — TUH's line drifts in amplitude within the 10 s fitting window, so a single stationary sinusoid does not describe it. After interpolation the 60 Hz region is flat in all four panels of Figure 8.1, while the notch leaves a deep dip in every cohort.

Examples: the signal itself

Figures 8.2–8.6 show what the numbers of Table 8.1 mean for real recordings. Two recordings with a clear 60 Hz line are taken as examples — the TUEG and the TUAB recording of the strong-line subset whose raw LNP_{60} is closest to 25 dB — and two clean ones (raw $|\mathrm{LNP}_{60}| < 1$ dB, typical amplitude). Each is shown raw (**before any filtering**; only the channel mean removed so the traces are centred) and after each protocol (**after processing**, 200 Hz): the LaBraM recipe (Eq. 1.1), the 60 Hz notch, the sinusoid fit (Eq. 8.1) and spectrum interpolation (Eq. 8.2). The traces show 6 of the 23 channels, 10 s from the 120 s segment of Section 3; the spectrograms use the whole segment.

Spectrogram. For channel c, the short-time power $S_c(f, \tau)$ (μV²/Hz) uses 2 s Hann windows with 1 s overlap (`scipy.signal.spectrogram`); the image shows the channel median in dB,

$$\mathrm{SG}(f, \tau) = 10 \log_{10} \operatorname{median}_c S_c(f, \tau) \qquad [\mathrm{dB\ re\ 1}\,\mu\mathrm{V}^2/\mathrm{Hz}] \tag{8.3}$$

with one colour scale per figure. A mains line is a horizontal bright stripe at 60 Hz; a notch hole is a horizontal dark stripe.

```
TUEG strong: aaaaasml_s001_t030  native 250 Hz, age 73, raw LNP60 24.8 dB
TUEG clean : aaaaaarg_s001_t001  native 400 Hz, age 36, raw LNP60 0.1 dB
TUAB strong: aaaaappy_s001_t000  native 250 Hz, age 55, raw LNP60 25.5 dB
TUAB clean : aaaaapxv_s001_t001  native 250 Hz, age 53, raw LNP60 0.0 dB
```

Figure 8.2. TUEG recording with a 60 Hz line: 10 s of EEG before and after each method

The TUEG strong-line example (stem and raw LNP_{60} printed above), 6 channels, seconds 20–30 of the 120 s segment, μV with a common trace spacing (red scale bar): (a) raw, before any filtering (native sampling rate, channel mean removed); (b)–(e) after processing with the LaBraM recipe, the 60 Hz notch, the sinusoid fit and spectrum interpolation (200 Hz). (f) One second (20–21 s) of channel C3 from (a)–(e) overlaid, where the 60 Hz

oscillation riding on the EEG is visible in the raw and LaBraM traces and gone after the other three.

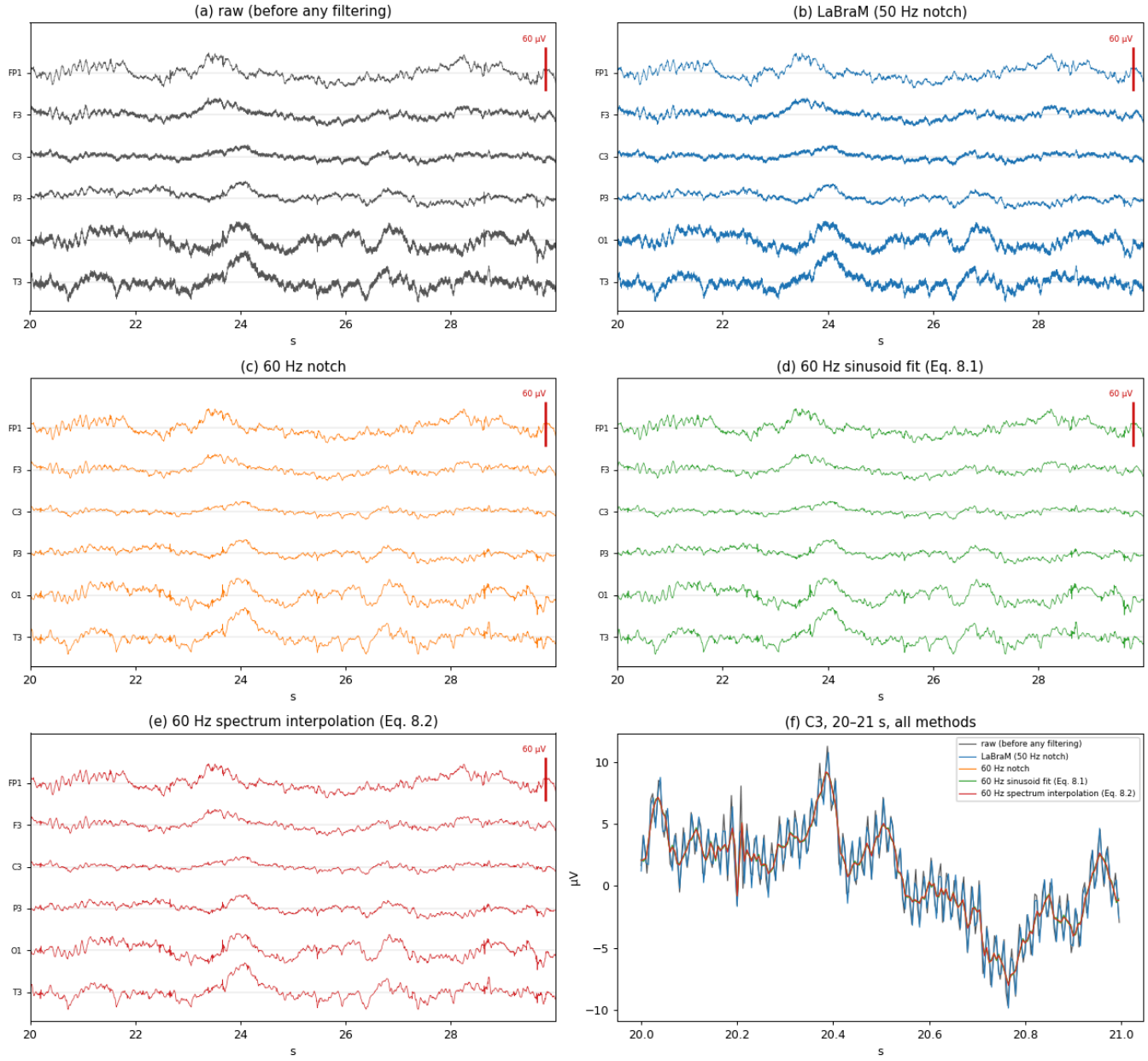


Figure 8.3. TUAB recording with a 60 Hz line: 10 s of EEG before and after each method

The TUAB strong-line example (stem and raw LNP_{60} printed above), 6 channels, seconds 20–30 of the 120 s segment, µV with a common trace spacing (red scale bar): (a) raw, before any filtering (native sampling rate, channel mean removed); (b)–(e) after processing with the LaBraM recipe, the 60 Hz notch, the sinusoid fit and spectrum interpolation (200 Hz). (f) One second (20–21 s) of channel C3 from (a)–(e) overlaid, where the 60 Hz oscillation riding on the EEG is visible in the raw and LaBraM traces and gone after the other three.

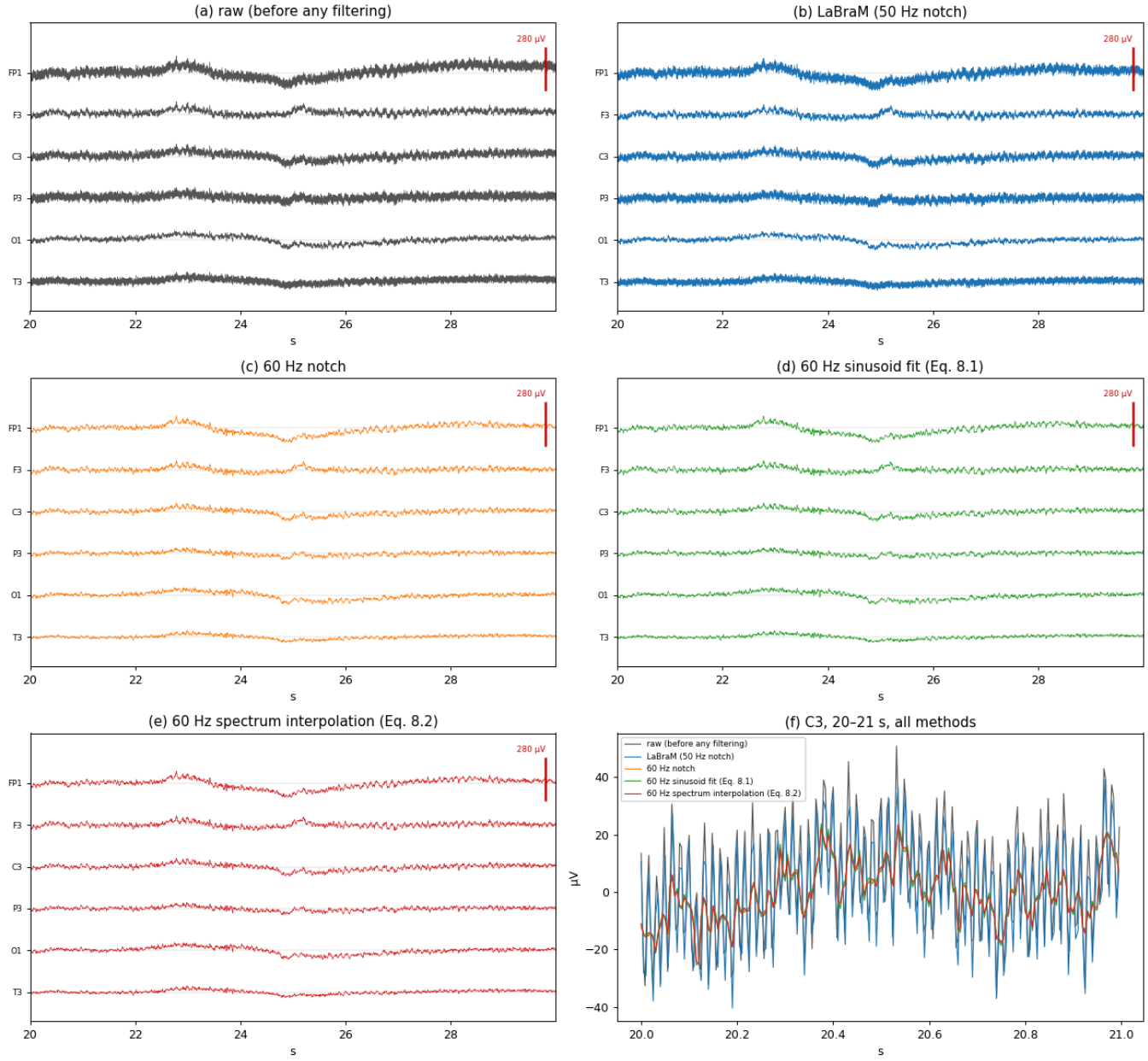


Figure 8.4. TUEG recording with a 60 Hz line: spectrogram before and after each method

The same TUEG recording as Figure 8.2, whole 120 s segment: (a) raw, before any filtering; (b)–(e) after processing with the LaBraM recipe, the 60 Hz notch, the sinusoid fit and spectrum interpolation. Colour: $\mathrm{SG}(f, \tau)$ of Eq. (8.3) in dB, one scale for all five panels; dashed line at 60 Hz. (f) The channel-median Welch spectrum (Eq. 3.1) of each version, 40–80 Hz.

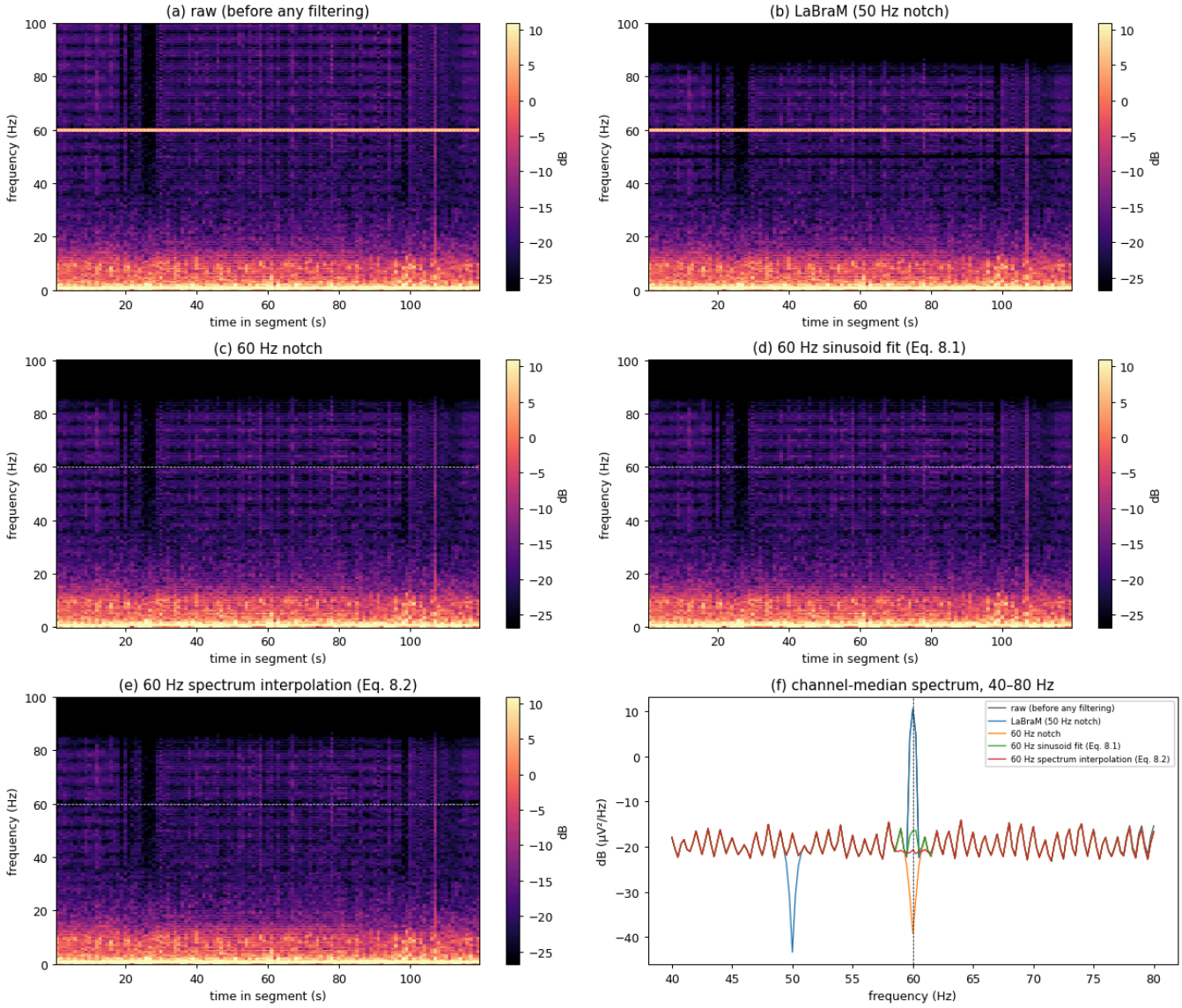


Figure 8.5. TUAB recording with a 60 Hz line: spectrogram before and after each method

The same TUAB recording as Figure 8.3, whole 120 s segment: (a) raw, before any filtering; (b)–(e) after processing with the LaBraM recipe, the 60 Hz notch, the sinusoid fit and spectrum interpolation. Colour: $\mathrm{SG}(f, \tau)$ of Eq. (8.3) in dB, one scale for all five panels; dashed line at 60 Hz. (f) The channel-median Welch spectrum (Eq. 3.1) of each version, 40–80 Hz.

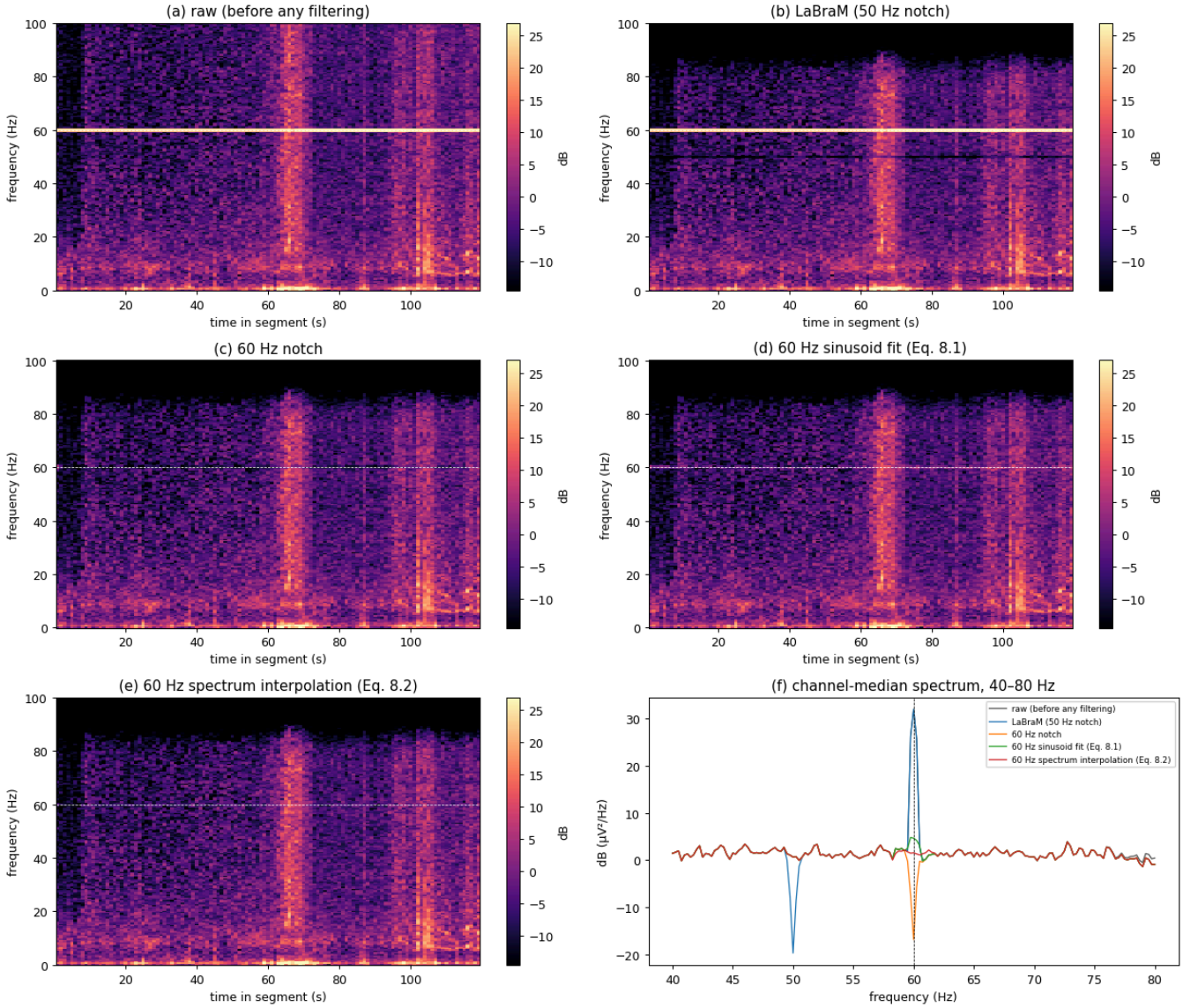
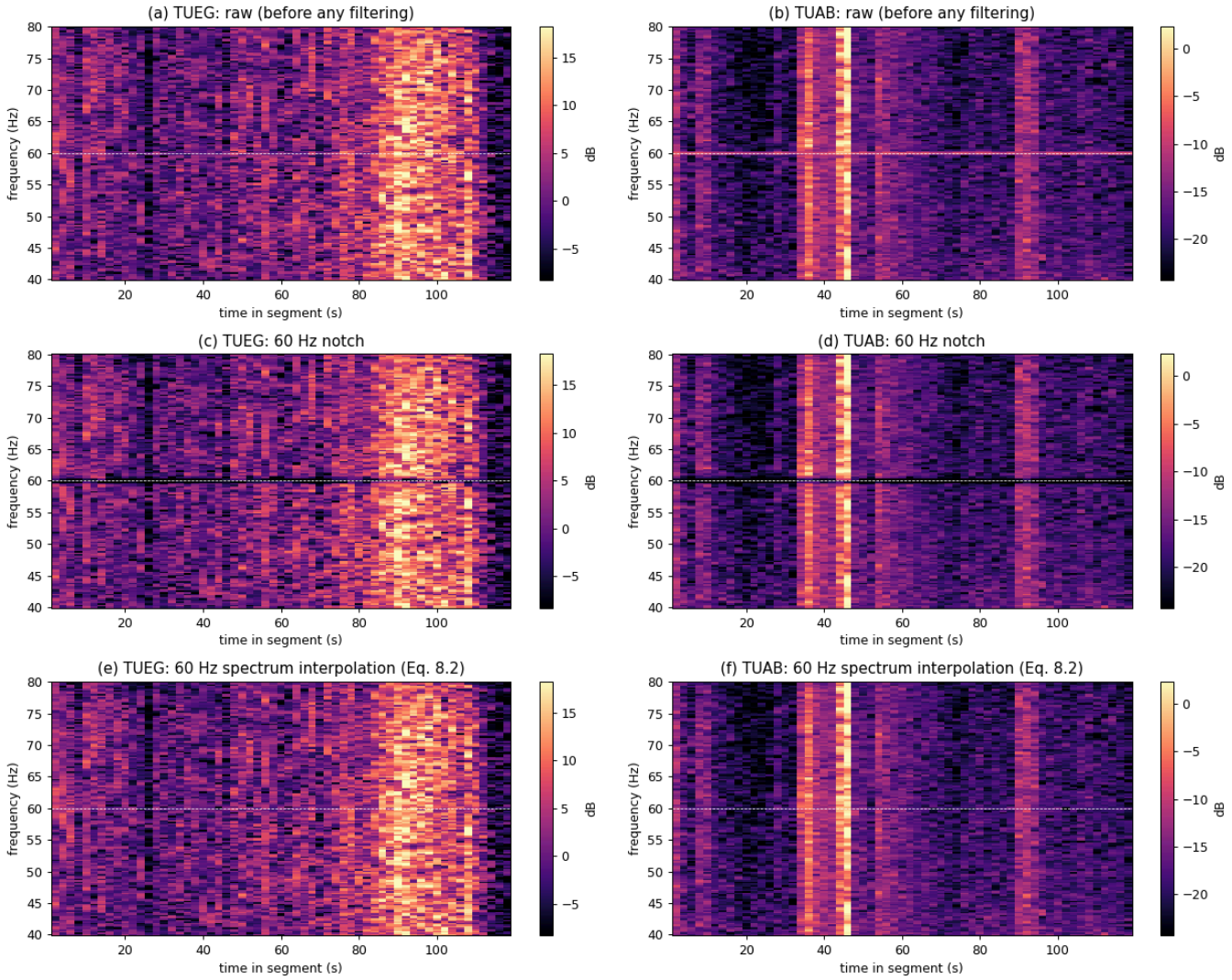


Figure 8.6. Clean recordings: the hole a notch cuts, and none from interpolation

The TUEG (left column) and TUAB (right column) clean examples (raw $|\mathrm{LNP}_{60}| < 1$ dB), spectrogram $\mathrm{SG}(f, \tau)$ (Eq. 8.3, here with 4 s windows for 0.25 Hz resolution, so a narrow hole is resolved) of the whole 120 s segment, zoomed to 40–80 Hz, one colour scale per column: (a, b) raw, before any filtering; (c, d) after the 60 Hz notch; (e, f) after spectrum interpolation. Dashed line at 60 Hz.



Figures 8.2 and 8.3 show the line as a fine, regular oscillation riding on every channel in the raw and LaBraM traces — most clearly in the one-second close-ups (Figures 8.2(f), 8.3(f)) — and gone after the 60 Hz notch and spectrum interpolation, with the EEG waveform otherwise unchanged. In the spectrograms (Figures 8.4, 8.5) the line is a bright horizontal stripe at 60 Hz through the whole segment in the raw and LaBraM panels; the notch replaces it with a dark stripe, the sinusoid fit leaves part of it where the line's amplitude drifts, and interpolation leaves the 60 Hz row indistinguishable from its neighbours. On clean recordings (Figure 8.6) the notch creates a dark stripe where there was nothing to remove, while interpolation leaves the spectrogram as it was.

One more feature of the TUEG example is unrelated to line removal: its spectrum carries a comb of ripples spaced exactly 1 Hz apart (Figure 8.4(f)), identical before and after every method. A periodic disturbance with a period of exactly 1 s — the length of one EDF data record in this file — produces such a comb. A separate check over the whole sample found it in only a few TUEG recordings and in none of TUAB; it belongs to per-recording quality control (detect and exclude), not to the choice of line-removal method.

What interpolation does where there is no line: 50 Hz on TUH

TUH has no 50 Hz interference (Table 4.1), so applying 50 Hz line removal to TUH shows directly what each method does to EEG **where there is nothing to remove** — the situation of any mains frequency that a corpus does not have. This matters for a single recipe across corpora: data from 50 Hz countries (e.g. hospital EEG recorded in Israel or Europe) has its line at 50 Hz, TUH at 60 Hz. Table 8.2 and Figures 8.7–8.8 compare, on all 187 recordings, band-pass only (the reference), the LaBraM 50 Hz notch, 50 Hz spectrum interpolation, 60 Hz interpolation, and the universal recipe of interpolating both 50 and 60 Hz. Besides LNP (Eq. 3.2) it reports the change of power in the 50 Hz band relative to band-pass only,

$$\Delta_{50} = 10 \log_{10} \frac{\sum_{|f-50|\le 1.5} P_{\text{method}}(f)}{\sum_{|f-50|\le 1.5} P_{\text{band-pass only}}(f)} \qquad [\mathrm{dB}] \tag{8.4}$$

per recording (0 dB = the method leaves the 50 Hz band unchanged).

Table 8.2. Line removal where there is no line (50 Hz) and where there is one (60 Hz)

All 187 recordings (TUEG + TUAB), after processing. `LNP50 med/min` (dB, Eq. 3.2): a negative value is a hole. `Δ50 med/min/max` (dB, Eq. 8.4): change of the 48.5–51.5 Hz power against band-pass only. `LNP60 med/max` (dB): 60 Hz line left. `%>6dB`: recordings with $\mathrm{LNP}_{60} > 6$ dB.

method	LNP50 med	LNP50 min	Δ50 med	Δ50 min	Δ50 max	LNP60 med	LNP60 max	%>6dB
band-pass only (no line removal)	0.0	-1.9	0.0	0.0	0.0	3.1	48.0	32.6
labram (50 Hz notch)	-5.1	-7.5	-1.3	-2.1	-0.8	3.1	48.0	32.6
50 Hz spectrum interpolation	-0.2	-1.3	-0.2	-1.6	1.7	3.1	48.0	32.6
60 Hz spectrum interpolation	0.0	-1.9	-0.0	-0.0	0.0	-0.3	2.2	0.0
50 + 60 Hz spectrum interpolation	-0.2	-1.3	-0.2	-1.6	1.7	-0.3	2.2	0.0
60 Hz notch	0.0	-1.9	0.0	-0.0	0.0	-5.3	7.4	1.1

Figure 8.7. 50 Hz line removal on data without a 50 Hz line

All 187 recordings (TUEG + TUAB), after processing. (a) Mean over recordings of $\log_{10} P(f)$ (μV²/Hz, Eq. 3.1), 40–60 Hz, for band-pass only, the LaBraM 50 Hz notch and 50 Hz spectrum interpolation; dashed line at 50 Hz. (b) Distribution over recordings of Δ_{50} (dB, Eq. 8.4) for the 50 Hz notch and 50 Hz interpolation; 0 dB = unchanged.

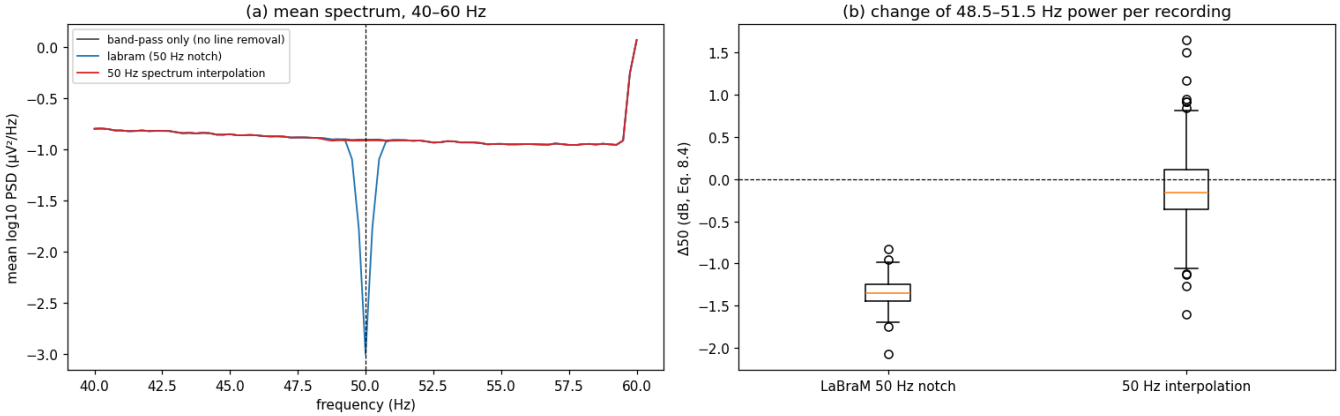
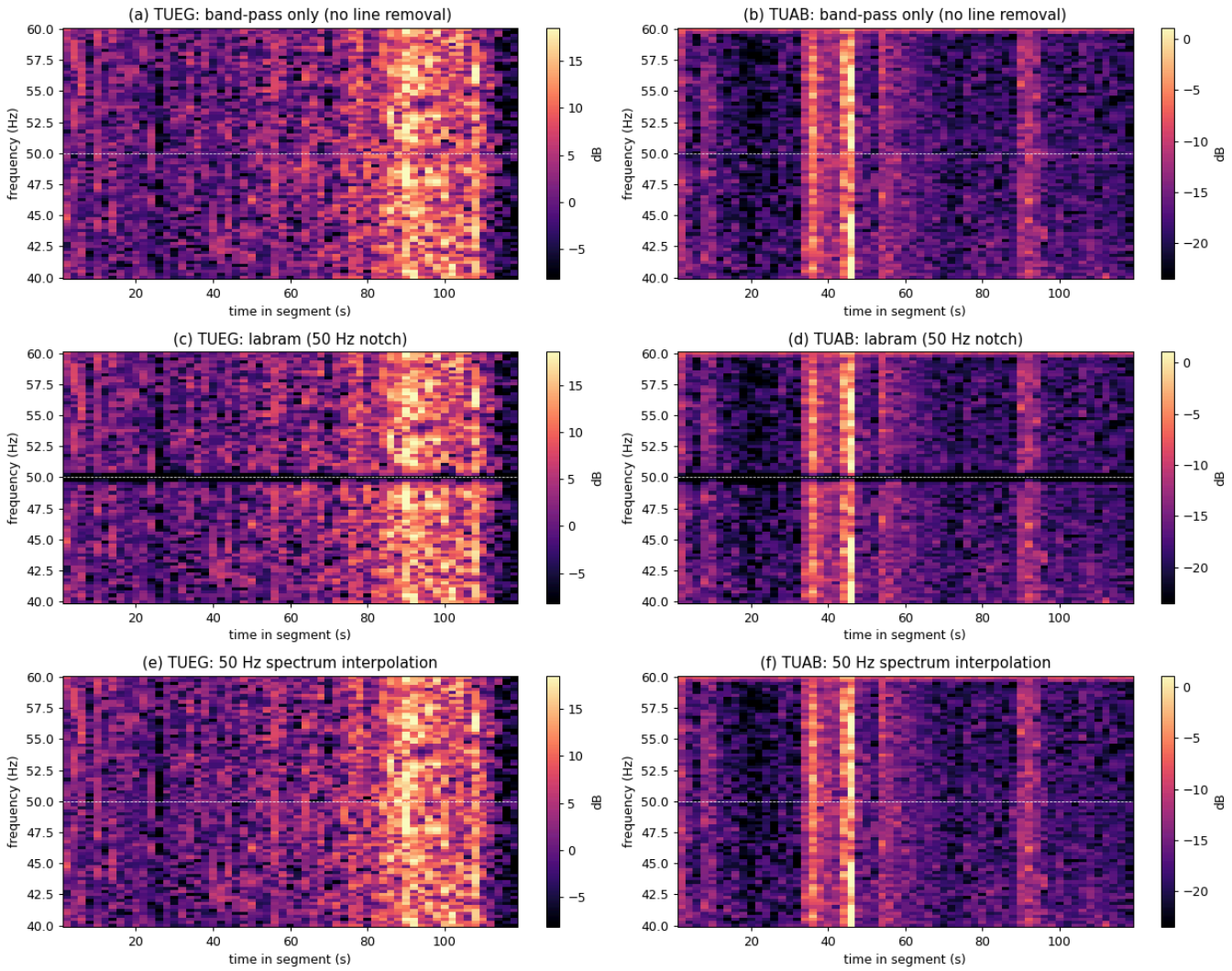


Figure 8.8. Clean recordings at 50 Hz: notch vs interpolation where there is no line

The clean TUEG (left) and TUAB (right) examples of Figure 8.6, spectrogram $\mathrm{SG}(f, \tau)$ (Eq. 8.3, 4 s windows) of the 120 s segment, 40–60 Hz, one colour scale per column: (a, b) band-pass only; (c, d) LaBraM 50 Hz notch; (e, f) 50 Hz spectrum interpolation. Dashed line at 50 Hz.



Where there is no line, **interpolation leaves the signal essentially unchanged**: the median Δ_{50} is a fraction of a dB and the mean spectrum is indistinguishable from band-pass only (Table 8.2, Figure 8.7(a)), while the LaBraM 50 Hz notch removes power from every recording (Figure 8.7(b)) and cuts a dark stripe into every spectrogram (Figure 8.8(c, d)) that is absent after interpolation (Figure 8.8(e, f)). The remaining spread of Δ_{50} for interpolation, of order ±1.5 dB, is the estimation noise of a single recording's 3 Hz band, centred on zero rather than a systematic loss. Two consequences:

- Interpolating at 50 Hz alone does **not** fix TUH: it leaves the 60 Hz line exactly as it was (Table 8.2, `LNP60` and `%>6dB` equal to band-pass only). TUH needs 60 Hz removal.
- Because interpolation is harmless where there is no line, **interpolating both 50 and 60 Hz** is a safe universal recipe: on TUH it gives the same result as 60 Hz interpolation alone (Table 8.2), and on 50 Hz data it would remove that line just as well. One recipe can therefore serve TUH and 50 Hz corpora mixed in one training set, with no corpus-specific preprocessing that a model could learn to tell apart.

9. Training evidence

Sections 4–8 measure the signal. This section asks whether the model notices: the same fine-tuning was run on three versions of TUAB v3.0.1 that differ **only** in line removal — **A** the 50 Hz notch of the LaBraM recipe (Eq. 1.1, the published preprocessing), **B** a 60 Hz notch, **C** spectrum interpolation at 50 and 60 Hz (Eq. 8.2) — built from the same EDFs by `dataset_maker/make_TUH_npy.py` (float32, identical recordings, windows and train/val/test split, `--split-from`). Two tasks:

- **brain age**: `labram/configs/defaults/finetune_tuab_age.json` (scenario D: common average reference, 60 s trims, 5-minute evaluation budget per recording);
- **normal vs abnormal**: `labram/configs/defaults/finetune_tuab.json` (the published TUAB setup).

Both on a short schedule (4 epochs, 1 warmup epoch; the published abnormal setup trains 50), from the pre-trained `labram-base` weights, with seeds 0 (local, one A10G) and 1–2 (SageMaker `ml.g5.2xlarge`, one A10G; `scripts/submit_line_removal_ab.sh`). The best epoch of each run is chosen on validation as the trainer does (lowest validation MAE for age, highest validation accuracy for abnormal) and the test metrics of that epoch are reported. Runs and metrics are collected by `scripts/collect_line_removal_ab.py` into `docs/data/line_removal_ab.json`.

Metrics. Predictions are pooled per recording (mean over its windows) before scoring. For recordings $r = 1..N$ with true age y_r and predicted age $\hat{y}_r$ (years),

$$\mathrm{MAE} = \frac{1}{N}\sum_{r=1}^{N}|\hat{y}_r - y_r| \qquad [\text{years}] \tag{9.1}$$

(`labram/utils/regression_metrics.py`), also within the normal and the abnormal recordings. For the abnormal task, with p_r the predicted probability of "abnormal" and the decision $p_r > 0.5$,

$$\mathrm{BAcc} = \tfrac{1}{2}\big(\mathrm{TPR} + \mathrm{TNR}\big) \tag{9.2}$$

(true-positive rate on abnormal, true-negative rate on normal recordings), and the threshold-free

$$\mathrm{AUROC} = \Pr\big(p_r > p_{r'} \,\big|\, r \text{ abnormal},\ r' \text{ normal}\big) \tag{9.3}$$

with AUPRC the area under the precision–recall curve. The `window` columns score every 10 s window on its own, which is how the LaBraM paper reports TUAB. Across seeds s, the effect of a variant V is the paired difference against A on the same seed,

$$\Delta_s(V) = m_s(V) - m_s(\mathrm{A}) \tag{9.4}$$

for a metric m; Table 9.3 gives its mean and its range over seeds, and in how many seeds V is better than A.

`18 runs: 3 seeds per task and variant`

Table 9.1. Brain age per line-removal variant

Best-on-validation epoch of each run; mean ± standard deviation over seeds (no ± with one seed). `val MAE` / `test MAE`: MAE (years, Eq. 9.1) on validation and test recordings; `normal` / `abnormal`: test MAE within normal and abnormal recordings; `r`: test Pearson correlation of predicted and true age; `seeds` = number of runs.

	val MAE	test MAE	normal	abnormal	r	seeds
A: 50 Hz notch	7.606 ± 0.073	8.357 ± 0.101	6.542 ± 0.114	10.520 ± 0.087	0.787 ± 0.002	3
B: 60 Hz notch	7.519 ± 0.049	8.130 ± 0.157	6.437 ± 0.095	10.148 ± 0.234	0.796 ± 0.005	3
C: 50+60 Hz interpolation	7.525 ± 0.058	8.146 ± 0.149	6.442 ± 0.090	10.179 ± 0.224	0.795 ± 0.004	3

Table 9.2. Normal vs abnormal per line-removal variant

Best-on-validation epoch; mean ± standard deviation over seeds. `val acc`: validation accuracy (the selection metric); `BAcc`, `AUROC`, `AUPRC`: test balanced accuracy (Eq. 9.2), AUROC (Eq. 9.3) and AUPRC per recording; `win BAcc`, `win AUROC`: the same per 10 s window (the LaBraM paper's convention); `seeds` = number of runs.

	val acc	BAcc	AUROC	AUPRC	win BAcc	win AUROC	seeds
A: 50 Hz notch	0.850 ± 0.007	0.847 ± 0.005	0.934 ± 0.003	0.929 ± 0.003	0.822 ± 0.007	0.899 ± 0.010	3
B: 60 Hz notch	0.841 ± 0.006	0.839 ± 0.006	0.933 ± 0.004	0.929 ± 0.006	0.819 ± 0.002	0.901 ± 0.004	3
C: 50+60 Hz interpolation	0.840 ± 0.006	0.859 ± 0.011	0.934 ± 0.008	0.932 ± 0.008	0.825 ± 0.005	0.902 ± 0.008	3

Table 9.3. Paired effect against the LaBraM recipe (A)

Δ_s of Eq. (9.4) per metric for B and C against A on the same seed: mean over seeds with [min, max]; `better` = seeds in which the variant beats A (lower MAE, higher accuracy/AUROC/AUPRC) out of the seeds run for both.

task	metric	variant	Δ mean	Δ min	Δ max	better
age	test MAE	B: 60 Hz notch	-0.227	-0.413	-0.093	3 / 3
		C: 50+60 Hz interpolation	-0.210	-0.399	-0.087	3 / 3
	test MAE normal	B: 60 Hz notch	-0.106	-0.231	-0.035	3 / 3
		C: 50+60 Hz interpolation	-0.101	-0.231	-0.018	3 / 3
	test MAE abnormal	B: 60 Hz notch	-0.372	-0.630	-0.144	3 / 3
		C: 50+60 Hz interpolation	-0.341	-0.600	-0.129	3 / 3
abn	BAcc	B: 60 Hz notch	-0.007	-0.009	-0.006	0 / 3
		C: 50+60 Hz interpolation	0.013	0.002	0.023	3 / 3
	AUROC	B: 60 Hz notch	-0.001	-0.006	0.004	1 / 3
		C: 50+60 Hz interpolation	-0.000	-0.009	0.008	2 / 3
	AUPRC	B: 60 Hz notch	0.000	-0.009	0.007	2 / 3
		C: 50+60 Hz interpolation	0.003	-0.009	0.011	2 / 3
	win AUROC	B: 60 Hz notch	0.002	-0.007	0.017	1 / 3
		C: 50+60 Hz interpolation	0.003	-0.011	0.021	1 / 3

Figure 9.1. Test metrics over training epochs

Every run as a thin line, the mean over seeds as a thick line, one colour per variant: (a) brain-age test MAE (years, Eq. 9.1) per epoch; (b) normal-vs-abnormal test AUROC per recording (Eq. 9.3) per epoch. Epochs 0–3 of the 4-epoch schedule.

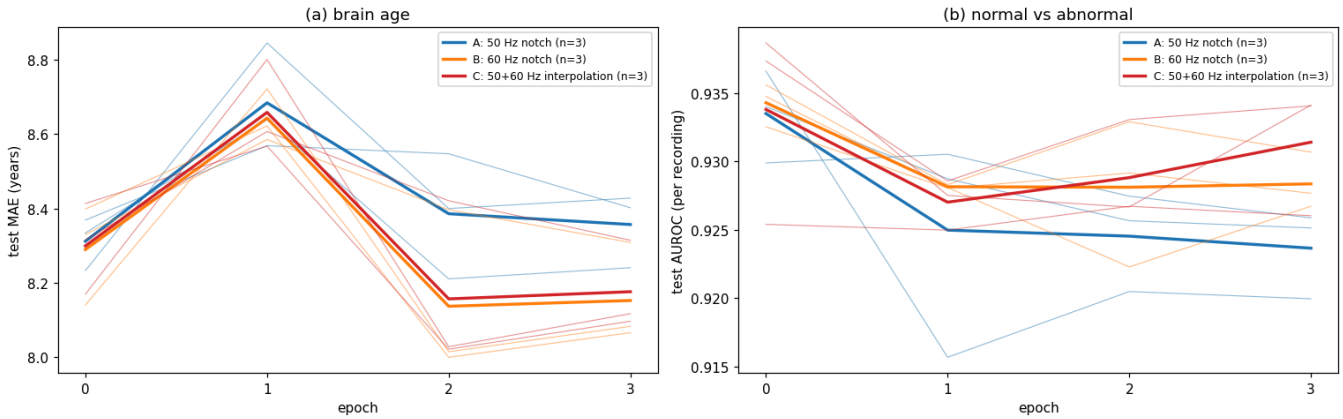
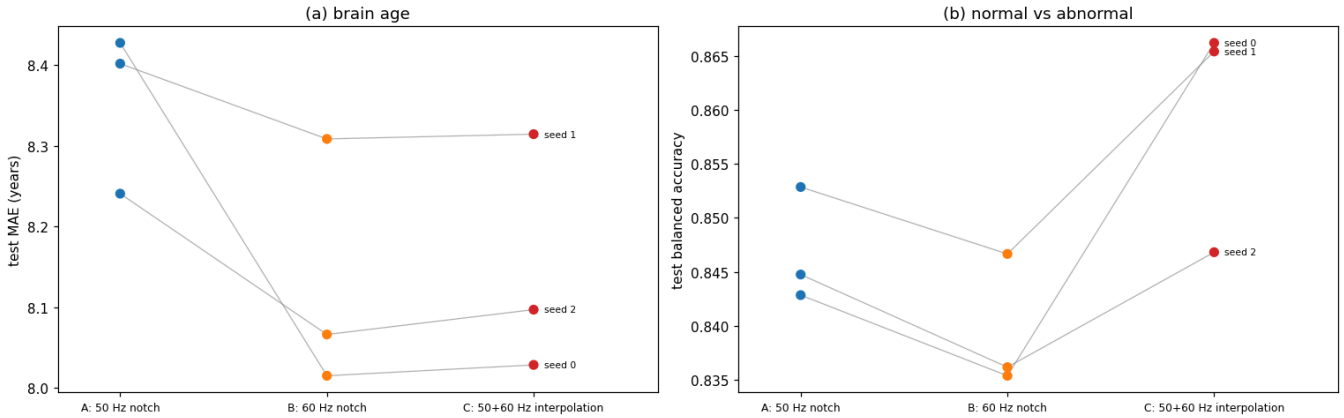


Figure 9.2. Per-seed results at the selected epoch

Each dot is one run at its best-on-validation epoch, lines join the three variants of the same seed: (a) brain-age test MAE (years, Eq. 9.1); (b) normal-vs-abnormal test balanced accuracy per recording (Eq. 9.2).



Brain age: removing the 60 Hz line helps, consistently but modestly. Both B and C lower the test MAE in all three seeds (Table 9.3, `better` 3 / 3): by about 0.2 years on average (B −0.23, C −0.21 years; per seed between −0.09 and −0.41), more on abnormal recordings (about −0.35 years) than on normal ones (about −0.10 years). Validation MAE falls as well (Table 9.1: about 7.61 for A against 7.52–7.53 for B and C), and on average over seeds B and C are below A at every epoch, by a few hundredths of a year at epochs 0–1 and about 0.2 years at epochs 2–3 (Figure 9.1(a)). The gain is about 2.5% of the MAE: small, but every seed agrees on its sign (Figure 9.2(a)).

Normal vs abnormal: no change in discrimination at the selected epoch. Test AUROC and AUPRC are the same for all three recipes within the seed spread (Table 9.2; Table 9.3: mean ΔAUROC about 0.00, ranges straddling zero). Balanced accuracy at the fixed 0.5 threshold moves in opposite directions — C higher than A in all three seeds (about +0.013), B lower in all three (about −0.007) — while the ranking of recordings (AUROC) does not change, i.e. a shift in calibration rather than in what the model can tell apart; validation accuracy is about 0.01 lower for B and C. The selected epoch is mostly the first; at the later epochs B and C have a slightly higher test AUROC than A on average (Figure 9.1(b)), a hint that only the full 50-epoch schedule can confirm. Line removal does not hurt the abnormal task.

Notch vs interpolation: equal. B and C differ by about 0.02 years of MAE and lie within each other's seed spread on every abnormal metric. The training results therefore do not choose between the two ways of removing the line; Section 8 does — interpolation cuts no hole and is safe where there is no line, so one 50 + 60 Hz recipe serves every corpus.

Limits: TUAB only, a short 4-epoch schedule, three seeds; the brain-age effect is consistent in sign but its size (≈0.2 years) is of the order of the seed-to-seed spread.

10. Conclusions and the recommended TUH protocol

1. **Mains: remove the line by spectrum interpolation at 50 and 60 Hz, not by a 50 Hz notch.** No recording has 50 Hz interference (Table 4.1); about half have a 60 Hz line

whose strength depends on the equipment generation, which is itself tied to recording year and patient age at group level (Tables 2.1, 6.2), though not to age recording by recording (Table 6.1). The LaBraM protocol passes this line to the model unchanged (Table 5.1, Figure 5.1(a, c)). Spectrum interpolation (Eq. 8.2, h = 1.5 Hz) removes it from every recording without the hole a notch cuts (Table 8.1, Figures 8.1–8.6), and leaves the signal unchanged where there is no line (Table 8.2, Figures 8.7–8.8) — so interpolating **both** 50 and 60 Hz is one safe recipe for TUH and for any 50 Hz corpus mixed with it. A 60 Hz notch is the fallback. Training confirms it (Section 9): removing the line lowers brain-age test MAE in every seed (about 0.2 years, Table 9.3), leaves the abnormal task's discrimination unchanged (Table 9.2), and notch and interpolation perform equally. TUEG v2.0.2 has been converted with this recipe (`processed_npy_interp50-60_float16/`, 60,386 recordings).
2. **Keep the 0.1–75 Hz band-pass and 200 Hz resampling.** They cause no aliasing at any native rate from 250 to 1000 Hz (alias probes in Tables 4.1 and 5.1) and match what the pre-trained LaBraM weights expect. A 45 Hz low-pass is not needed once the line is removed.
3. **Units are consistent** (all µV, Table 2.2); the header prefiltering field is uninformative, so per-recording QC must come from the signal.
4. **Amplitude differs by corpus and equipment** (Table 7.1, Figure 7.1); keep CAR, and compare per-patch z-scoring against fixed scaling in the first TUEG experiments.
5. **Consistency:** TUAB must be preprocessed with exactly the same recipe as TUEG when the two are mixed. For the TUAB benchmark (comparison with published numbers) the original 50 Hz recipe stays available.

Table 10.1 summarises the recommendation.

Table 10.1. LaBraM protocol vs the recommended TUH protocol

One row per preprocessing step; `evidence` names the table or figure that supports the change.

step	LaBraM (TUAB papers)	recommended for TUH	evidence
channels	23 referential, exact set required	23 referential, missing ones zero-filled + channel_mask	corpus index (23% of TUEG recordings miss ≥1 channel)
band-pass	0.1–75 Hz	0.1–75 Hz (unchanged)	Tables 4.1, 5.1
mains removal	50 Hz notch	spectrum interpolation at 50 and 60 Hz (±1.5 Hz); 60 Hz notch as fallback	Tables 4.1, 5.1, 6.2, 8.1, 8.2, 9.1–9.3; Figures 5.1, 8.1–8.8, 9.1–9.2
resample	200 Hz	200 Hz (unchanged)	Table 5.1 (no aliasing)
units / scale	µV, model input x/100	µV; CAR on; z-score vs fixed scaling tested	Table 7.1, Figure 7.1
TUAB	as published (50 Hz notch)	re-made with the same recipe as TUEG (50 + 60 Hz interpolation); published recipe kept only for benchmark comparison	Section 10, item 5